

Problem 1 - Washing Machine Mount Design (1 DOF)

Problem 1a

The Matlab code for this problem is listed in Appendix 1.

Figure 1 shows the load force produced by the 1 kg imbalanced mass at a 0.5 m radius from the center of the washing machine as a function of angular speed in RPM.

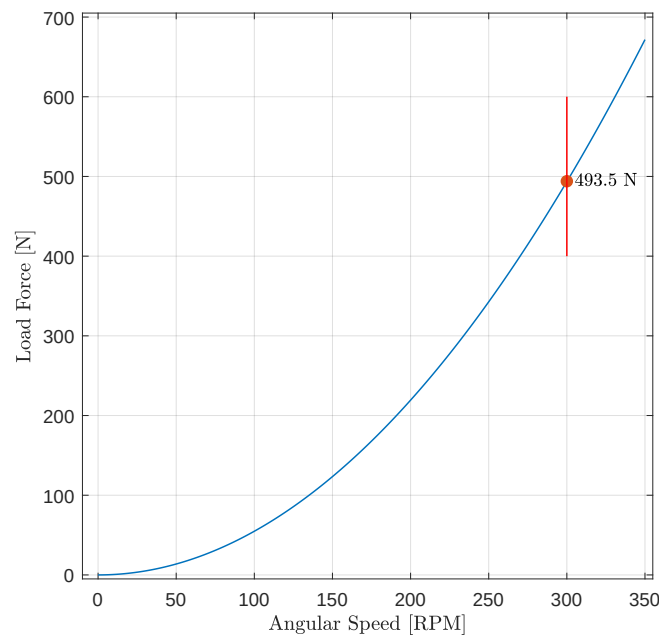


Figure 1: Load force of 1 kg imbalanced mass at a 0.5 m radius versus washing machine angular speed.

The force amplitude is 493.5 N at a frequency of 5 Hz when the washing machine operates at 300 RPM.

Problem 1b

The static displacement of the load in the washing machine must be less than 1 cm (i.e., the deflection is calculated with only the 10 kg load mass). The corresponding mount stiffness, k_s , is **9,810 $\frac{\text{N}}{\text{m}}$** .

The natural frequency, ω_o , of the mount is **1.5 Hz** (or $9.4 \frac{\text{radians}}{\text{s}}$).

Problem 1c

Figure 2 shows the transmission loss of the system for a set of damping ratios ranging from 0.001 to 1.

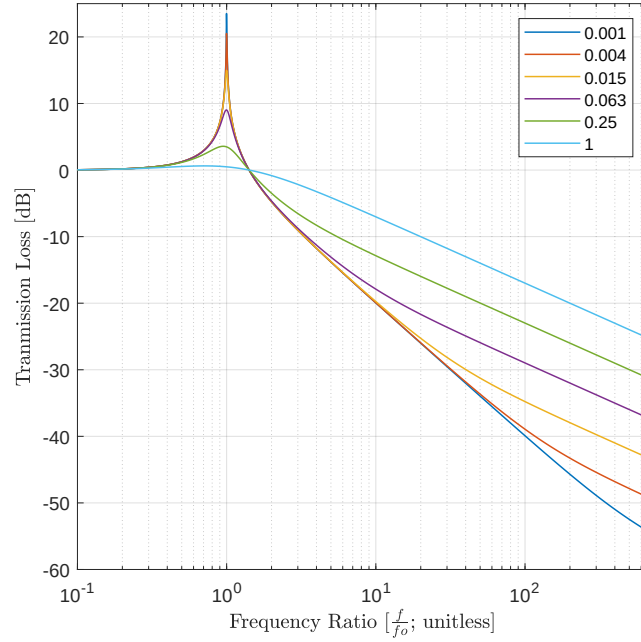


Figure 2: Transmission loss of the system. Its natural frequency is 1.5 Hz.

Problem 1d

Figure 3 shows the force applied to the foundation as a function of RPM.

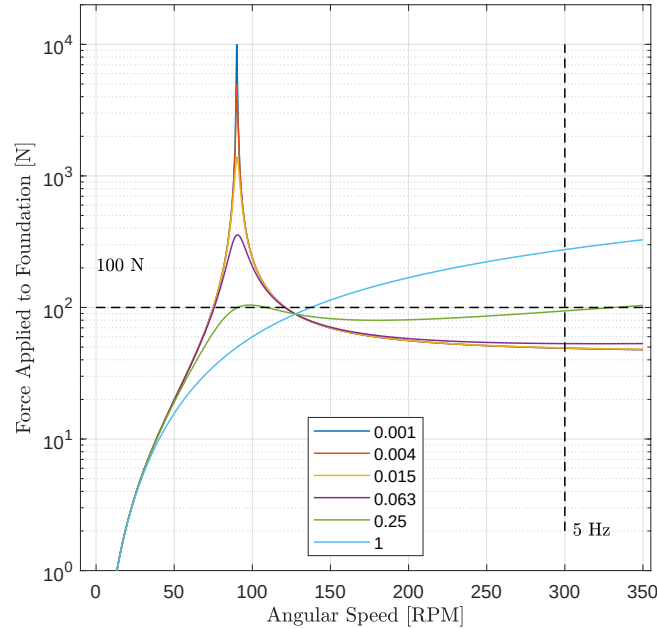


Figure 3: Force applied to the foundation versus RPM.

There is no damping ratio for which the applied force is less than 100 N across the whole operating range. The best damping ratio is $\zeta = 0.25$. With this damping ratio, the force applied to the foundation still exceeds the 100 N target in two regions: 91 to 109 RPM and 331 to 350 RPM. In both regions the force is about 104 N.

Problem 1e

Figure 4 shows the admittance as a function of RPM.

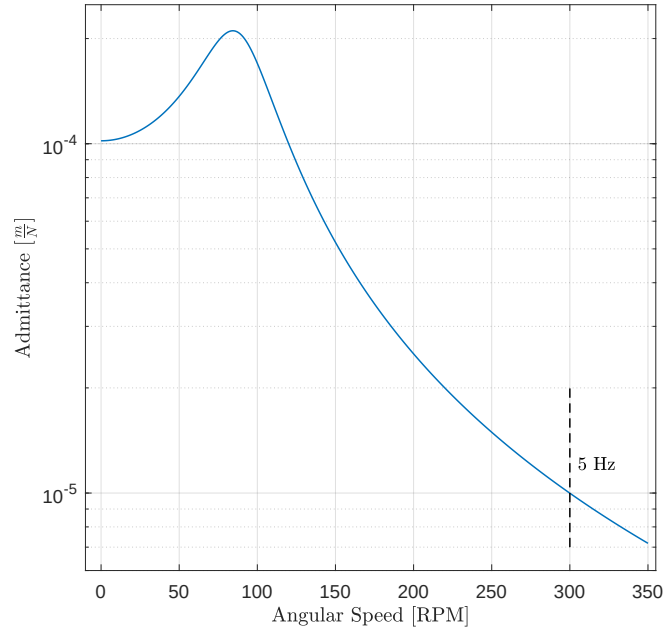


Figure 4: Admittance versus RPM.

The admittance at 300 RPM is $10^{-5} \frac{m}{N}$.

Problem 1f

Figure 5 shows the displacement of the washing machine as a function of RPM.

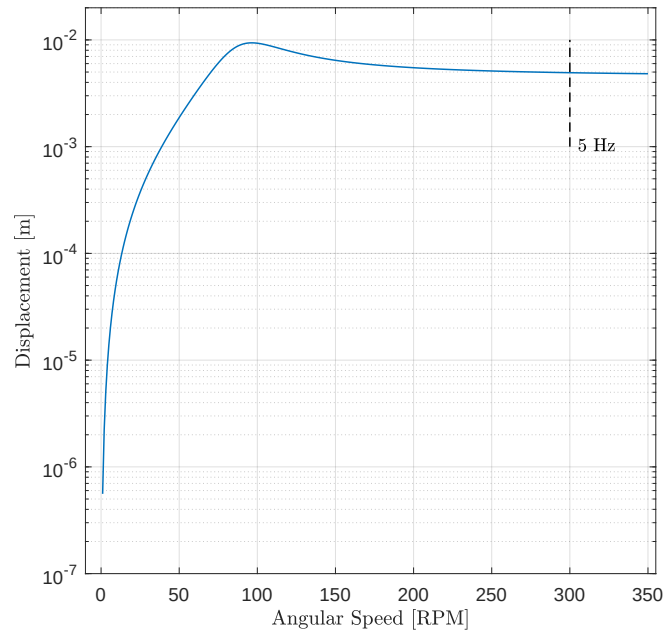


Figure 5: Displacement versus RPM.

The displacement of the washing machine at 300 RPM is 5×10^{-3} m. This amount of displacement for the 300 RPM operating state seems excessive. The washing machine undergoes larger momentary displacements as it spins up to the 300 RPM operating state, in particular around 100 RPM.

Problem 2 - Washing Machine Mount Design (2 DOF)

Problem 2a

The Matlab code for this problem is listed in Appendix 2.

Figure 6 illustrates the two-stage vibration mount considered here (see slide 8 of 42 for Lecture 14 on Monday, March 3, 2025). Modeling will be done using discrete viscous damping elements.

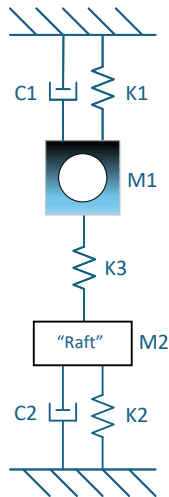


Figure 6: Two-stage vibration mount system for the washing machine.

Problem 2b

Table 1 lists the initial stiffness and damping coefficients for the system.

M1	110	kg
K1	100	$\frac{\text{N}}{\text{m}}$
C1	5	$\frac{\text{kg}}{\text{s}}$

M2	100	kg
K2	100	$\frac{\text{N}}{\text{m}}$
C2	5	$\frac{\text{kg}}{\text{s}}$

M3	0	kg
K3	1,000	$\frac{\text{N}}{\text{m}}$
C3	5	$\frac{\text{kg}}{\text{s}}$

Table 1: Initial values of stiffness and damping coefficients.

Problem 2c

Figure 7 illustrates the admittance of the washing machine and the raft with the force applied to the washing machine.

At the operating speed of 300 RPM, the admittance of the washing machine is $9.31 \times 10^{-6} \frac{\text{m}}{\text{N}}$.

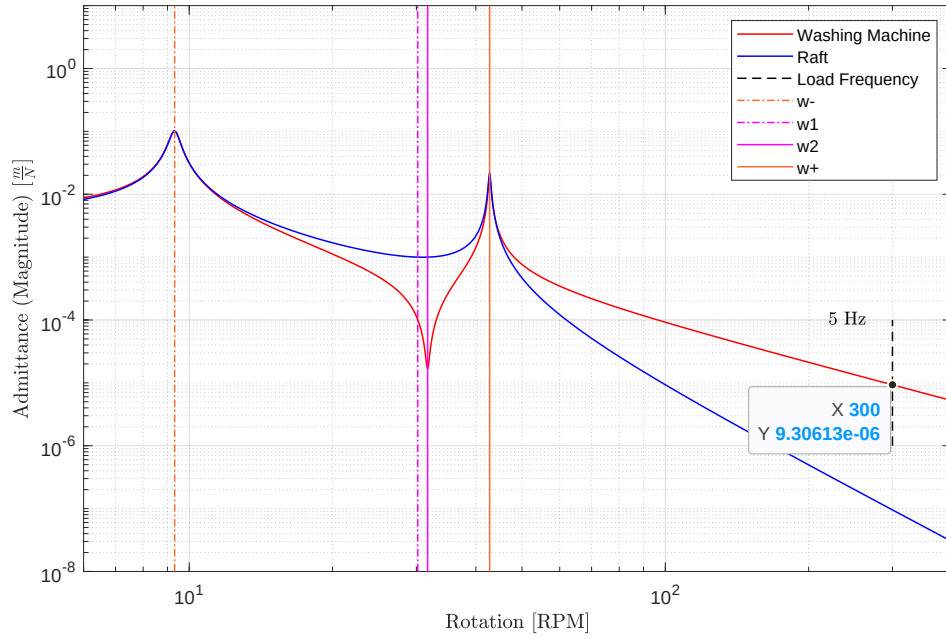


Figure 7: Admittance of the washing machine and the raft with a force on the washing machine.

Problem 2d

Figure 8 illustrates the displacement of the washing machine and the raft with the force applied to the washing machine.

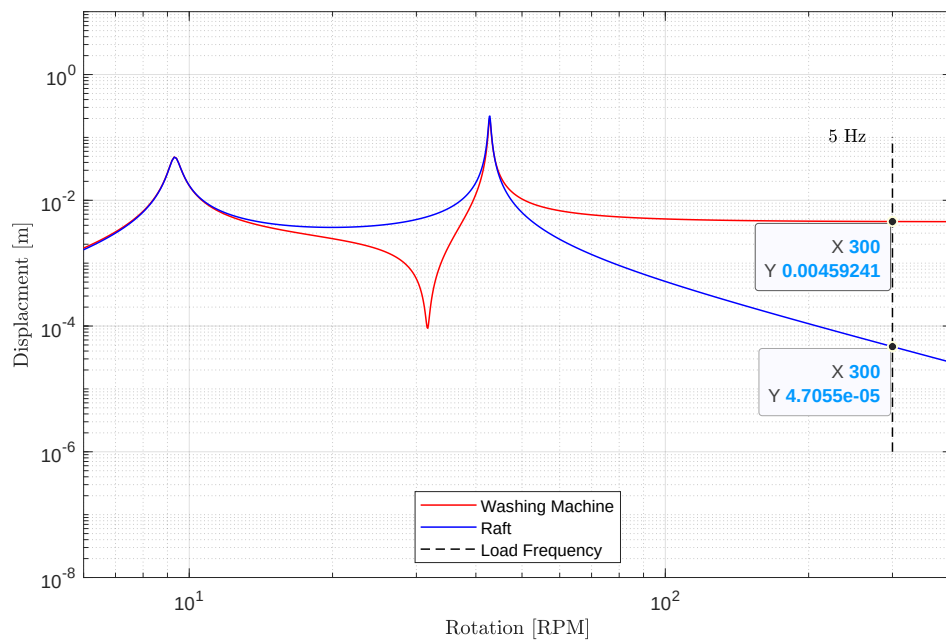


Figure 8: Displacement of the washing machine and the raft with a force on the washing machine.

At the operating speed of 300 RPM, the displacement of the washing machine is 4.6×10^{-3} m, which is less than the 5.0×10^{-3} m displacement with the first-order system.

The displacement of the raft is 4.7×10^{-5} m.

Problem 2e

Figure 9 illustrates the force applied to the ground by the raft. This is calculated by multiplying the displacement of the raft by the K2 spring constant.

The load on the ground is always less than 100 N.

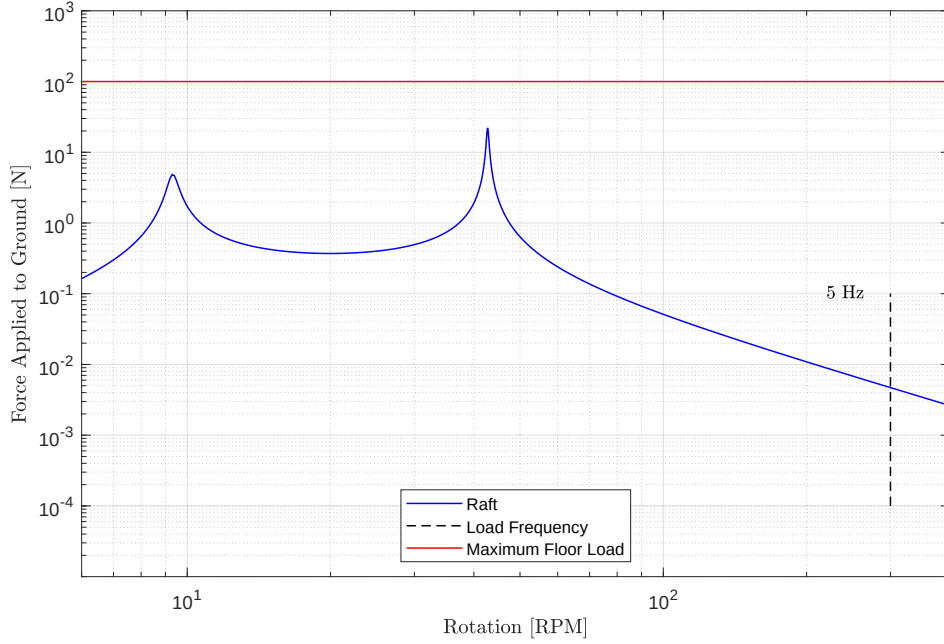


Figure 9: Force applied to the ground by the raft.

Problem 2f

The initial parameter set chosen, see Table 1, facilitated a displacement of the washing machine of 4.6×10^{-3} m at 300 RPM, which is less than the 5.0×10^{-3} m found for the 1 DOF system. The force applied to the ground by the raft is less than the required 100 N maximum.

Problem 2g

The initial parameter set chosen, see Table 1, facilitated a displacement of the washing machine of 4.6×10^{-3} m at 300 RPM, which is less than the 5.0×10^{-3} m found for the 1 DOF system. The force applied to the ground by the raft is less than the required 100 N maximum.

Problem 3 - Washing Machine Dynamic Vibration Absorber Design

Equations from DVA Tutorial Lecture

The Matlab code for this problem is listed in Appendix 3.

A DVA is used at a particular operating frequency, here 300 RPM or 5 Hz. Not used across a wide range of operating RPMs.

The ramp up to 300 RPM occurs quickly so that the large displacements are transient.

Higher roll-off of admittance with a DVA system, relative to the system in Problem 1 (see video at -09:35 mark).

Adding damping to the DVA system reduces the prominence of its two resonance peaks, while still reducing the amount of admittance and displacement of the original mass.

Figure 10 ph

ph

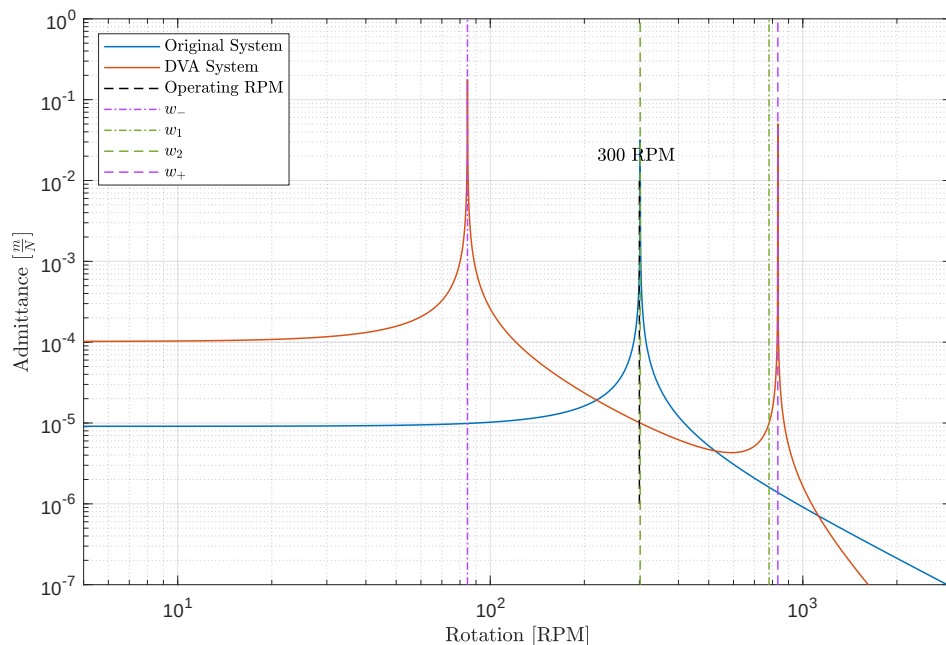


Figure 10: Equations from DVA tutorial lecture.

Figure 11 ph

ph

Figure 12 ph

ph

Figure 13 ph

ph

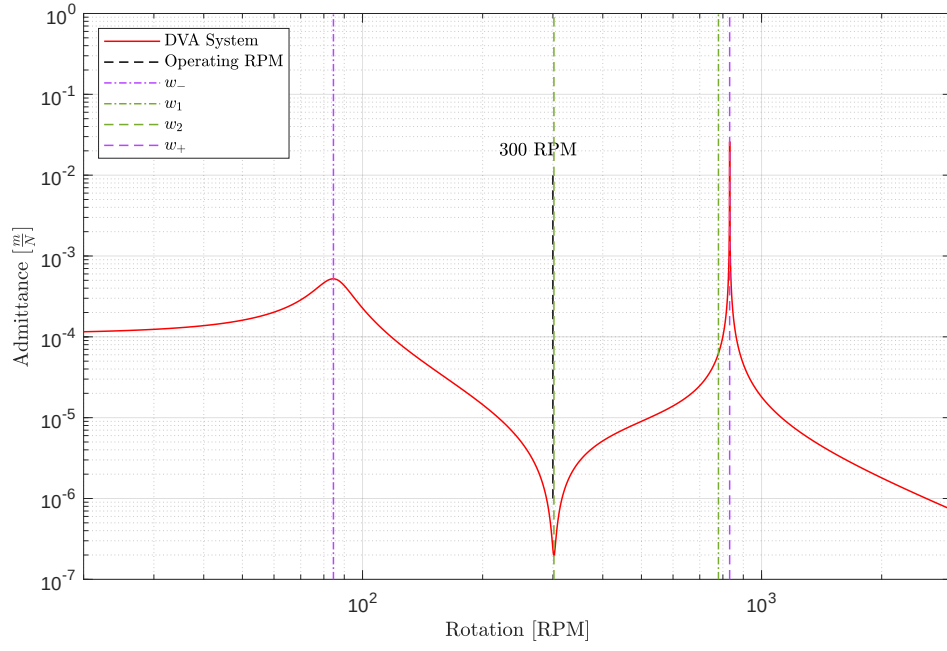


Figure 11: Solution using nDOF script-file function.

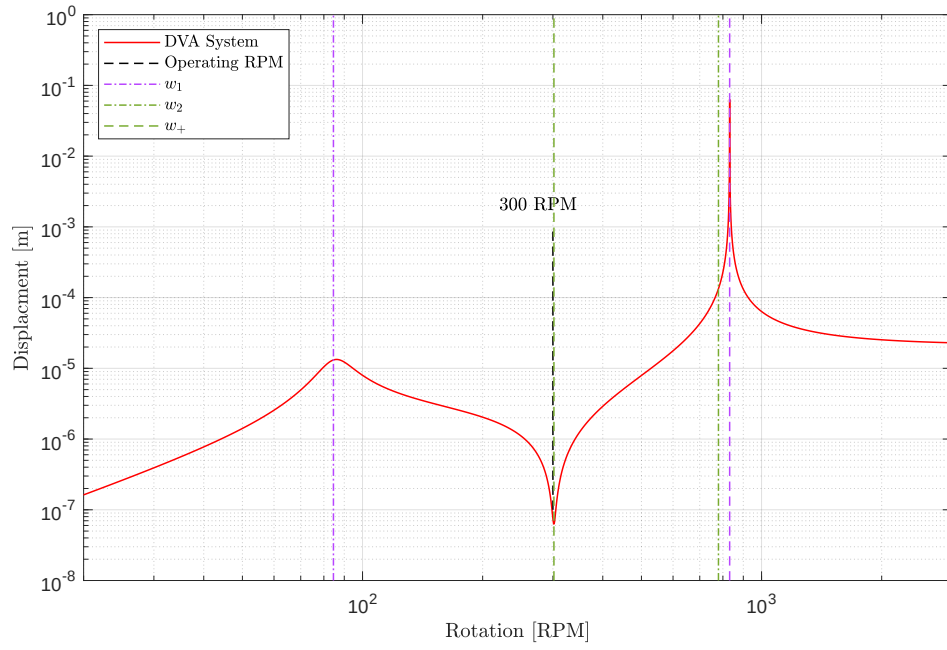


Figure 12: Displacement of the washing machine.

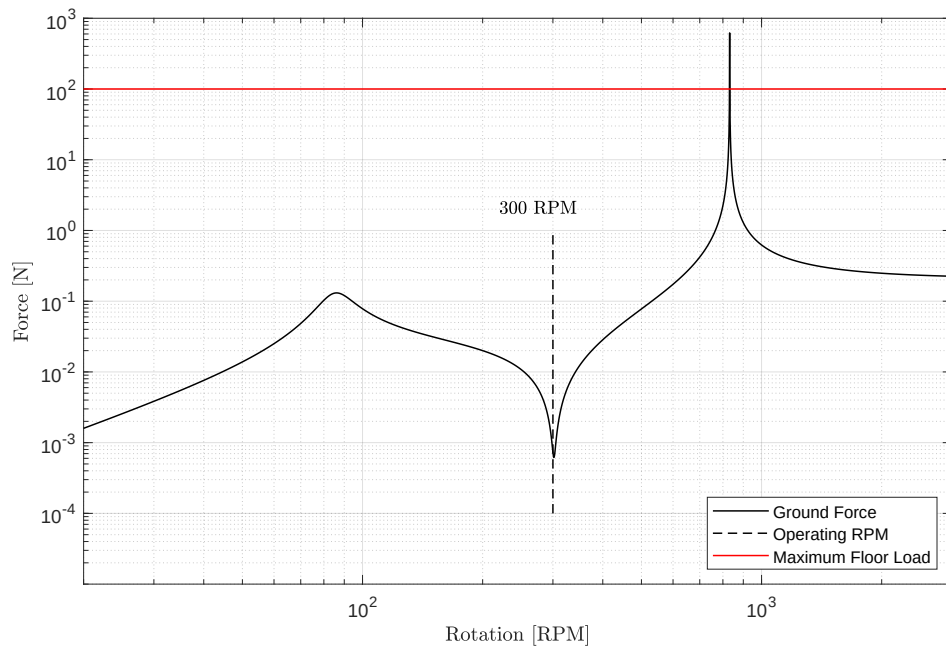


Figure 13: Force applied to the floor.

1 Appendix - Matlab Code for Problem 1

2 Appendix - Matlab Code for Problem 2

3 Appendix - Matlab Code for Problem 3

4 Appendix - Matlab Code for Problem 4